Lightweight Post-training for Controllable Vietnamese Summarization

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Abstract

Controllable Vietnamese summarization requires a model to preserve source content while also obeying explicit output budgets. We present an empirical study of a lightweight post-training pipeline that combines instruction-conditioned LoRA supervised fine-tuning (SFT) with a custom group relative policy optimization (GRPO) loop for two Qwen3-4B backbones. The main paper evaluates the resulting post-trained systems on four datasets (VietNews, WikiLingua, ViMs, and VLSP) and situates them against contextual baselines such as VietAI, GPT-4o, GPT-3.5-Turbo, Qwen3-14B, and Llama3.3-70B-Instruct using ROUGE-2 and absolute summary length distance. On the two single-document datasets that support our strongest claims, the final `SFT + GRPO v5` systems substantially improve over their raw Qwen3 backbones and compare favorably to several historical, non-protocol-matched external rows while keeping low length distance; the Instruct system reaches 29.62 ROUGE-2 on VietNews, and the Base system reaches 26.35 on WikiLingua. We keep ViMs and VLSP visible in the benchmark tables as multi-document diagnostics whose current test pools overlap with project training or validation data. Their weaker and less consistent performance is plausibly explained by the current training setup: multi-document supervision is extremely scarce, contributes only minimally to the dominant SFT stage, and is represented through flat document-cluster concatenation rather than explicit cross-document modeling. Targeted ablations further show that SFT is the dominant improvement stage, that warm-starting GRPO from the SFT checkpoint is better than applying GRPO directly to the pretrained backbone, and that the safeguarded reward design sharply reduces failure-prone behavior relative to the earlier GRPO setup. The code for the project is available at https://github.com/tranphuc15122004/PoML_for_summary.

1 Introduction

Controllable summarization is useful when systems must meet strict output budgets, such as headline generation, short mobile previews, or template-bound reporting. In these settings, a model must preserve salient content while also satisfying an explicit length target. That combination is still difficult for lightweight open-weight language models, especially in Vietnamese, where controllable summarization resources and post-training analyses remain limited.

This paper studies a practical post-training recipe for that setting as an empirical case study of lightweight Vietnamese controllable summarization. Our focus is not to enumerate model variants for their own sake, but to explain a method (instruction-conditioned LoRA-SFT followed by custom GRPO) and to position its final outputs against external baselines on the datasets currently available in the project. The two instantiated project systems are the Base and Instruct backbones after SFT followed by GRPO.

We evaluate the final systems on four datasets and emphasize two main metrics: ROUGE-2 for content overlap and absolute summary length distance for control accuracy. VietNews and WikiLingua provide the cleanest single-document evidence. ViMs and VLSP remain informative for multi-document behavior, but their current test sets overlap with project training or validation pools and therefore appear in the paper with explicit caveats about split overlap. Their weaker results are also methodologically unsurprising in the present setup: the training distribution is overwhelmingly single-document, the dominant SFT stage sees very little multi-document supervision, and document clusters are serialized as flat prompts rather than modeled with dedicated cross-document structure.

The paper makes three empirical contributions. First, it documents a lightweight Vietnamese controllable summarization pipeline from prompt construction through LoRA-SFT, GRPO alignment, and evaluation. Second, it reports a four-dataset benchmark that situates the final systems alongside historical baselines such as VietAI, GPT-4o, GPT-3.5-Turbo, and Qwen3-14B while keeping protocol caveats visible. Third, it includes focused analyses of the training stages, of the failure-prone behavior observed under earlier GRPO settings, and of the current gap between single-document and multi-document behavior. The appendix preserves the supporting evidence behind these claims: the full within-project stage tables, the historical configuration-bundle comparison, the multi-document baseline comparison, the sentence-control results, and the sentence-reward ablation.

2 Related Work

Vietnamese summarization benchmarks. Vietnamese summarization has moved from benchmark construction toward stronger modeling, but the literature is still comparatively small. VNDS established an early Vietnamese news summarization benchmark, ViMs expanded the area with a high-quality multi-document abstractive corpus, WikiLingua added a multilingual how-to benchmark that includes Vietnamese, and the VLSP AbMuSu shared task provided another multi-document news setting for evaluation (**????**). Modeling papers then showed that pretrained Vietnamese seq2seq backbones matter: ViT5 improved Vietnamese abstractive summarization, and VieSum/BRIO reported further gains from BRIO-style training on a Vietnamese summarization corpus (**??**). In this line of work, however, the main objective is still generic summary quality, not explicit control over length or sentence budget.

Controllable and length-aware summarization. Controllable summarization has progressed from architectural length conditioning to prompt-based instruction following. Early work controlled output length through decoder modifications or positional encodings, and later work generalized control to keywords or natural-language prompts (**?????**). More recent systems tighten the length problem further by targeting exact token or sentence counts and by benchmarking instruction-controllable summarization directly (**??**). Prompt-based multi-control methods and instruction-based length control show that LLMs can follow structured control text, but they also show that precise adherence is still fragile, especially outside English or under multiple simultaneous constraints (**??**). This project sits in that gap: it studies reference-conditioned Vietnamese control with coupled length and sentence targets rather than free-form user preference prompting.

Post-training for summarization behavior. Our training recipe belongs to the broader post-training line rather than to architecture changes. LoRA makes adaptation practical by freezing the backbone and learning low-rank adapters, which is attractive when the desired behavior is narrow and the model is large (**?**). Summarization work with learned rewards has also shown that optimizing a better reward signal can outperform ROUGE-only training, as in learning from human feedback (**??**). Recent policy-optimization methods such as GRPO further reduce the memory cost of on-policy alignment and have become common in open-model post-training; in that sense, our custom rollout-reward loop is best viewed as a lightweight adaptation of this broader family rather than a new optimizer (**?**). The contribution here is the application of these ideas to controllable Vietnamese summarization, not a new post-training algorithm.

Reward design and reward hacking. Once a model is optimized against a proxy reward, the reward itself becomes a design object. The AI safety literature uses the terms reward hacking and specification gaming for cases where optimizing the proxy produces superficially successful but substantively bad behavior (**??**). In summarization, reward-model work has long shown that ROUGE is an imperfect proxy for human judgment, and later analyses of reward-model overoptimization make the same point more sharply: increasing proxy reward can stop correlating with true quality (**?**). This literature motivates the content gate, degenerate-output filter, and multiplicative reward structure used in our GRPO setup, and it also explains why we treat the changes between the pre-safeguard and safeguarded configurations as a package-level mitigation rather than a single-component ablation.

Dataset	Task	Test	Role	Caveat
VietNews	Single-document news	2,000	Main	Single-doc. evidence; small duplicate caveat
WikiLingua	Single-document how-to	500	Main	Single-doc. evidence
ViMs	Multi-document news clusters	300	Multi	Test overlaps 240 train and 60 val. clusters
VLSP	Multi-document summarization	300	Multi	Test overlaps 285 train and 15 val. items

Table 1: Datasets used in the report. VietNews and WikiLingua provide the strongest single-document evidence. ViMs and VLSP remain visible in the benchmark tables but have split overlap with project training or validation data.

3 Datasets

3.1 Task and Prompt Setting

Each example consists of a source document or document cluster x, a control instruction c, and a reference summary y^*. The control instruction specifies a target length and sentence budget derived from the reference summary statistics, so the task is best viewed as *reference-conditioned control* rather than arbitrary user-defined control. At inference time, the model receives the source and the control instruction and produces $\hat{y}$.

The instruction schema is shared across SFT, GRPO, and evaluation. All prompts are built by the project's data augmentation pipeline. Word counts are computed as whitespace-delimited tokens, and sentence counts are computed from end punctuation. SFT always uses the approximate constraint template family, GRPO rotates across approximate, range, and upper-bound templates to diversify the control language, and evaluation fixes the approximate template family for deterministic testing against the frozen artefacts.

3.2 Dataset Inventory and Validity Boundary

Table 1 summarizes the four datasets used in the report. VietNews and WikiLingua are the only single-document datasets without known split overlap and therefore support the main generalization claims. ViMs and VLSP remain in the main benchmark because they reveal how the final systems behave on multi-document inputs, but they are explicitly marked for split overlap throughout the paper.

After preprocessing, SFT uses 111,150 samples: 96,911 VietNews items whose titles contain at least 10 whitespace-delimited words, 13,999 WikiLingua items, and 240 ViMs clusters. GRPO uses a broader 119,942-sample pool: all 105,418 VietNews training items, the same 13,999 WikiLingua items, 285 VLSP items, and the same 240 ViMs clusters. Appendix A.5 records the exact preprocessing rules and Appendix A.6 shows compact prompt examples.

4 Method

Figure 1 presents the proposed lightweight post-training pipeline for controllable Vietnamese summarization. The upper row follows the end-to-end flow from task and data through preprocessing and prompt construction, Stage 1 instruction-conditioned LoRA-SFT, Stage 2 GRPO alignment, and the resulting post-trained models. The lower panel expands the GRPO training loop and the safeguard package that constrains reward optimization. At a high level, the method first teaches a Qwen3-4B backbone to summarize under explicit control instructions, then aligns that behavior with grouped rollout optimization under content-aware rewards and anti-reward-hacking defenses. The following subsections mirror this structure and make the algorithmic contributions explicit.

4.1 Task Formulation, Preprocessing, and Prompt Construction

Each training or evaluation instance is represented as a triplet (x, c, y^*), where x is a source document or document cluster, c is a control instruction, and y^* is a reference summary. The control instruction is not arbitrary user input; it is derived from y^* after preprocessing, which truncates source and summary fields to the working context budget and computes target word and sentence counts from the reference. The task is therefore controllable summarization under reference-conditioned budgets rather than unconstrained free-form prompting.

The pipeline constructs chat-style prompts rather than plain source-target pairs. For SFT, each example contains system, user, and assistant turns, and the control language is instantiated with approximate templates such as "around X words." For GRPO and evaluation, the assistant response is removed, the reference is stored separately for reward computation or scoring, and the

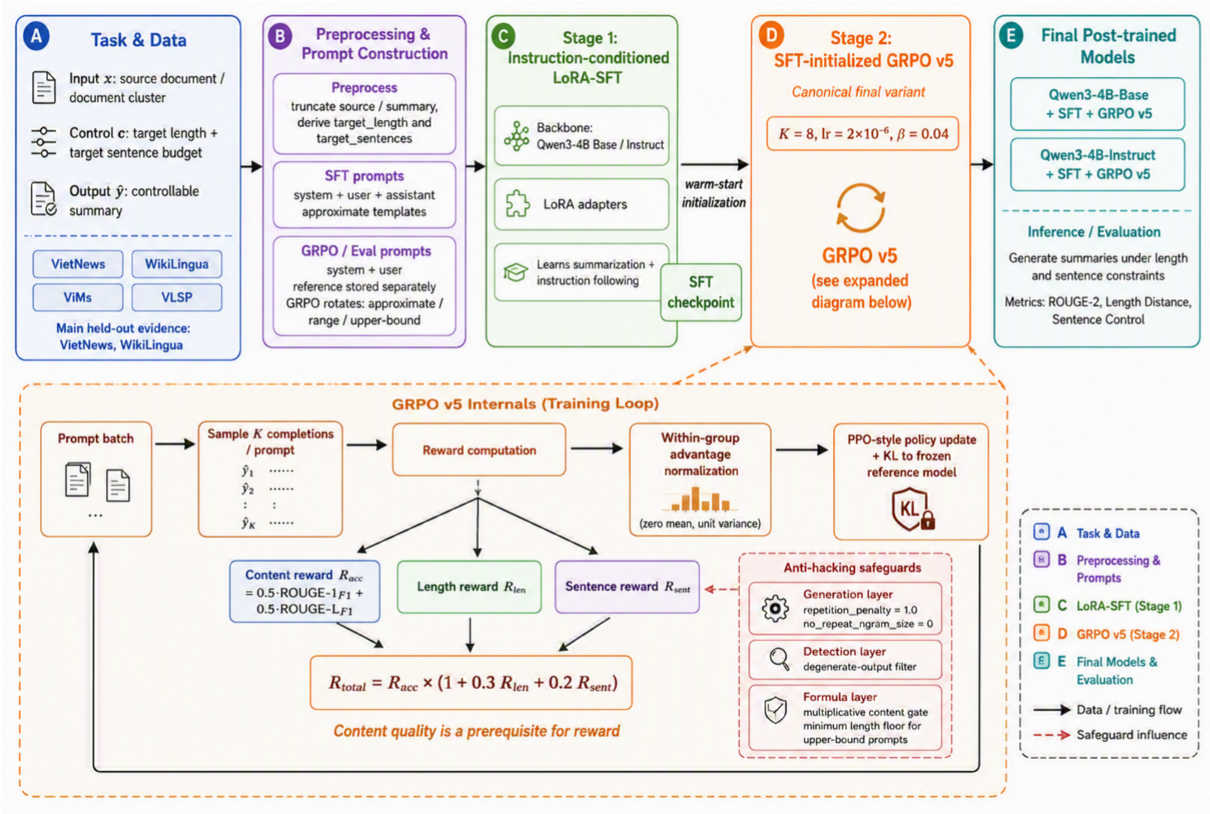

Figure 1: Overview of the proposed lightweight post-training pipeline for controllable Vietnamese summarization. The upper row summarizes task and data, preprocessing and prompt construction, Stage 1 instruction- conditioned LoRA-SFT, Stage 2 GRPO alignment, and the resulting post-trained models. The lower panel expands the GRPO training loop: prompt batches are decoded into multiple completions, rewarded for content and control adherence, normalized within each prompt group, and updated with a clipped policy objective plus KL regularization to a frozen reference model. The safeguard block summarizes the generation, detection, and formula layers used to reduce reward-hacking behavior.

model must generate a summary that respects the requested control budget from the system and user turns alone. Across GRPO and evaluation, the control language rotates across three template families: approximate constraints, explicit ranges, and upper bounds. This asymmetric prompt construction is deliberate: Stage 1 teaches the summarization task under a stable instruction schema, while Stage 2 exposes the policy to broader control wording without changing the underlying supervision target.

4.2 Stage 1: Instruction-conditioned LoRA-SFT

Stage 1 gives the backbone its base summarization and instruction-following behavior. Each processed $\{source, target\}$ pair is converted into an instruction-following chat example, so the model learns not only to compress Vietnamese content but also to map explicit control instructions to the desired output form. In practice, this stage is responsible for the largest single improvement in instruction following and summarization quality in the project, and it produces the checkpoint used to warm-start the GRPO stage.

Both Qwen3-4B families studied in the paper, Base and Instruct, are adapted with LoRA over the attention and MLP projection layers. All canonical runs use LoRA with rank $r = 32$, scaling parameter $\alpha = 32$, and dropout 0.05. The main SFT runs use a 3,072-token context length, learning rate 5×10^{-5}, warmup ratio 0.1, two epochs, and thinking mode suppression enabled.

4.3 Stage 2: GRPO Alignment

Stage 2 is implemented as a custom rollout-and-reward loop rather than through a stock trainer abstraction. In each training step, the policy formats a batch of prompts, samples multiple completions per prompt, scores them with the reward model, normalizes the rewards within each prompt group, and applies a PPO-style policy update with

KL regularization. The frozen reference model is the *initial GRPO policy*: for *Fresh* GRPO it is the pretrained backbone plus a newly attached LoRA adapter at initialization, whereas for *SFT-initialized* GRPO it is the backbone plus the SFT checkpoint. The KL term therefore anchors optimization to the exact starting policy rather than to the raw backbone.

The project studies two initialization modes. *Fresh* GRPO attaches a new LoRA adapter directly to the pretrained backbone, while *SFT-initialized* GRPO warm-starts from the SFT checkpoint. In the final paper narrative, fresh GRPO is treated as an ablation baseline and SFT-initialized GRPO is treated as the intended deployment recipe.

Within each update, the trainer formats a batch of prompts, samples K completions per prompt, computes rewards $r_{i,k}$, and normalizes them within each prompt group:

$$A_{i,k} = \frac{r_{i,k} - \bar{r}_i}{\max(\sigma_i, 0.01)}, \tag{1}$$

where i indexes prompts and k indexes the sampled completions for prompt i. The policy is then updated with a PPO-style clipped objective over generated tokens plus a KL penalty to the frozen reference model. The per-token probability ratio is defined as

$$\rho_{i,k,t} = \exp\big(\log \pi_\theta - \log \pi_{\text{old}}\big), \tag{2}$$

and the policy objective with the clipped surrogate is

$$c_{i,k,t} = \text{clip}(\rho_{i,k,t}, 1-\epsilon, 1+\epsilon), \tag{3}$$

$$L_{\text{policy}} = -\frac{1}{|M|} \sum_{i,k,t} \min\big(\rho_{i,k,t} A_{i,k},\ c_{i,k,t} A_{i,k}\big) m_{i,k,t}, \tag{4}$$

with the total loss including a KL penalty:

$$L = L_{\text{policy}} + \beta L_{\text{KL}}, \tag{5}$$

where $m_{i,k,t}$ masks generated tokens up to the first EOS token. In the implementation, π_{old} is the detached current policy within the same on-policy step, and the KL term is computed token-wise against the frozen reference model. Among the stored configurations, the final systems instantiate this stage with the canonical `v5` bundle, which uses $K = 8$, learning rate 2×10^{-6}, and $\beta = 0.04$. This makes the method a lightweight GRPO-style rollout-and-update loop tailored to the present summarization setting rather than a wrapper around a stock trainer, with Stage 2 acting as a control-refinement and failure-mitigation layer on top of the capabilities learned in Stage 1.

4.4 Reward Design and Anti-Hacking Safeguards

The reward combines content quality with explicit control adherence. Content quality is measured as

$$R_{\text{acc}} = 0.5\, \text{ROUGE-1}_{F1} + 0.5\, \text{ROUGE-L}_{F1}. \tag{6}$$

Length and sentence rewards, denoted R_{len} and R_{sent}, are computed from the same instruction-specific tolerance rules used to construct the control prompts in the previous subsection. The final reward uses a multiplicative gate:

$$R_{\text{total}} = R_{\text{acc}} \times (1 + 0.3 R_{\text{len}} + 0.2 R_{\text{sent}}), \tag{7}$$

which makes content quality a hard prerequisite: when $R_{\text{acc}} = 0$, the total reward is zero regardless of constraint satisfaction.

Because constraint-based rewards are susceptible to reward hacking, where a model learns to satisfy length and sentence budgets with garbage content, the reward function is embedded in a defense-in-depth strategy with three layers. At the **generation** layer, the repetition penalty is kept at 1.0 and n-gram blocking is disabled (size 0). Preliminary experiments showed that higher values of either parameter, when applied by the HuggingFace decoder over the full $\approx$ 2,000-token input sequence, penalized the model for reusing source-article vocabulary and pushed decoding toward out-of-distribution symbols, which is especially damaging for short Vietnamese summaries where named entities and key terms must appear verbatim.

At the **detection** layer, a post-hoc degenerate-output filter examines each rollout with five sequential checks: empty or whitespace-only output; single-token blobs longer than 25 characters without spaces; fewer than three alphabetic characters; alpha character ratio below 12% when exceeding 15 characters; and token repetition where one token accounts for over 60% of all tokens or unique tokens fall below 30%. Any output flagged by this filter receives $R_{\text{len}} = 0$ and $R_{\text{sent}} = 0$, so degenerate rollouts cannot earn any constraint bonus; under the multiplicative gate, only residual content overlap captured by R_{acc} can remain.

At the **formula** layer, the multiplicative gate structurally prevents constraint-only optimization: no amount of length or sentence adherence can compensate for zero content quality. Additionally, the upper-bound constraint "khong qua X tu" enforces a floor of $\max(3, 0.15 \times \text{max\_wc})$ words; outputs shorter than this floor receive $R_{\text{len}} = 0$, closing the loophole of satisfying the constraint with one- or two-word garbage.

4.5 GRPO Variant Nomenclature and Canonical Configuration

The paper uses a compact nomenclature for the GRPO method variants along two orthogonal axes: initialization path and configuration bundle. *Fresh* GRPO applies GRPO directly to a pretrained backbone without an SFT stage, whereas *SFT-initialized* GRPO warm-starts GRPO from the corresponding SFT checkpoint. Table 2 formalizes the compact labels used throughout the remainder of the paper.

Label	Defining properties	Paper role
v3 (pre-safeguard)	Linear reward; repetition penalty 1.3; n-gram blocking size 3	Reward-hacking failure baseline
v4 (historical bundle)	$K = 4$, learning rate 5×10^{-7}, $\beta = 0.15$	Bundle comparison and sentence-reward ablation
v5 (canonical bundle)	$K = 8$, learning rate 2×10^{-6}, $\beta = 0.04$; gated reward and safeguards	Main paper configuration

Table 2: Compact nomenclature for the GRPO method variants used in the paper. Each label denotes a configuration bundle and, where applicable, a safeguard package; the labels should not be interpreted as variants differing in only one hyperparameter.

The canonical final systems instantiate the SFT-initialized `v5` branch, use 800 training steps, and keep the same LoRA adapter shape as SFT. The two final project systems differ only in their underlying Qwen3-4B backbone family: Base or Instruct. Earlier GRPO settings are retained as named historical comparison points rather than alternative deployment recipes. Appendix A.6 records the exact prompt templates and launch-script presets used to instantiate the canonical runs, while Appendix A.2 preserves the historical v4-v5 bundle comparison.

5 Experimental Results

5.1 Experimental Protocol

All tables in this paper are filled from frozen project artefacts rather than reruns. The main project evaluation, the v3 comparison, and the no-sentence ablation are drawn from three frozen evaluation snapshots produced at different stages of the project. Evaluation uses temperature 0.3, top-p 0.9, random sampling, and a maximum of 256 new tokens; no evaluation seed was saved. The main paper reports only ROUGE-2 and absolute summary length distance. Sentence-control results, relative length error, dataset-processing details, and prompt examples are moved to the appendix, where Appendix A.4 keeps the sentence-control table, Appendix A.7 records the no-sentence ablation, and Appendix A.1 preserves the full within-project variant tables.

The external baseline rows in Tables 3 and 4 come from a historical benchmark collection maintained alongside the project. They are useful contextual comparisons, but they are not protocol-matched reruns under the frozen evaluation pipeline used by the project systems. Accordingly, VietNews and WikiLingua remain the strongest evidence for model comparison inside the present report, while ViMs and VLSP are read as multi-document diagnostics; Appendix A.3 retains the full baseline-relative comparison for those datasets.

5.2 Main Benchmark Against External Baselines

Tables 3 and 4 provide the headline benchmark of the paper. The two project rows are the final `SFT + GRPO v5` systems, and the remaining rows are comparison baselines for Section 5.2.

Figure 2 summarizes the same benchmark in terms of percentage improvement of the two final systems over each baseline. Each dataset appears explicitly as a two-column block (ΔR2 and LenRed), so the figure complements Tables 3 and 4 without replacing their raw per-dataset values.

The two main tables show the raw benchmark values, while Figure 2 makes the comparison direction explicit. On VietNews and WikiLingua, both final systems improve strongly over the raw

Model	VietNews	WikiLingua	ViMs	VLSP
Project systems				
Qwen3-4B-Base + SFT + GRPO v5	28.97	<u>26.35</u>	18.16	17.46
Qwen3-4B-Instruct + SFT + GRPO v5	<u>29.62</u>	25.59	21.27	24.39
Raw Qwen3 baselines (frozen project protocol)				
Qwen3-4B-Base (raw)	12.13	3.18	13.46	9.49
Qwen3-4B-Instruct (raw)	16.48	7.44	17.83	15.51
Historical external baselines				
VietAI	**34.24**	**33.12**	–	–
GPT-4o	21.61	20.65	<u>44.26</u>	<u>43.37</u>
GPT-3.5-Turbo	28.13	21.09	19.28	35.79
Qwen3-14B	18.51	20.24	**44.38**	**44.27**
Llama3.3-70B-Instruct	22.17	19.40	37.54	39.02
Phi4-14B	13.17	14.28	41.85	42.31
Sailor-20B-chat	17.70	18.74	39.47	40.96
VinBigdata (7B)	20.59	21.02	37.98	40.23

Table 3: ROUGE-2 (%) across the four report datasets. Bold marks the best reported value in each column and underlining marks the second-best reported value across both project and baseline rows. The two final systems and the raw Qwen3 baseline rows are evaluated under the frozen project protocol; the remaining external rows are drawn from a historical benchmark collection and should be read as contextual rather than protocol-matched reruns. ViMs and VLSP have split overlap.

Qwen3 baselines and compare favorably to several external rows with available length-control values; VietAI remains the strongest ROUGE-2 row on both datasets, and the external baselines should still be read as contextual because they are not protocol-matched reruns. The same figure also shows that the method is not merely beating unrelated baselines: relative to the raw Qwen3 backbones, both final systems make especially large ROUGE-2 gains on WikiLingua and strong length-distance reductions on both single-document datasets.

ViMs and VLSP remain visible rather than hidden. Figure 2 shows that both final systems still improve over the raw Qwen3 baselines on ViMs and VLSP ROUGE-2, but they remain substantially below the strongest external multi-document baselines such as GPT-4o and Qwen3-14B on that metric. The ViMs/VLSP length-control picture is mixed rather than uniformly positive, especially for the Instruct branch against stronger baselines. This weaker multi-document picture is consistent with several properties of the current pipeline, although the evidence is observational rather than causal. First, as described in Section 3, SFT sees only 240 ViMs clusters out of 111,150 training examples and no VLSP at all, while GRPO adds only 525 multi-document prompts out of 119,942. Second, because Section 5.3 shows that SFT is the dominant gain stage, the lack of substantial multi-document supervision in that stage is likely consequential. Third, the pipeline serializes document clusters as flat concatenations of document blocks under the same prompt schema used for single documents, without an explicit document-level fusion mechanism. Finally, the current reward remains single-reference and ROUGE-oriented, which may under-reward valid multi-document summaries whose cross-document content selection differs from the one gold reference. Taken together, these factors provide a plausible explanation for why the method transfers more cleanly to the single-document setting than to the present multi-document diagnostics. Appendix Table 11 retains the full baseline-relative percentage comparison for readers who want the detailed ViMs/VLSP view.

Between the two backbone instantiations, the Instruct branch is the stronger content model overall, winning ROUGE-2 on VietNews, ViMs, and VLSP. The Base branch is slightly stronger on WikiLingua ROUGE-2 and also yields the best single-document length distance, especially on WikiLingua. This is why the paper keeps both fi-

Model	VietNews	WikiLingua	ViMs	VLSP
Project systems				
Qwen3-4B-Base + SFT + GRPO v5	**1.16**	**8.35**	89.23	100.50
Qwen3-4B-Instruct + SFT + GRPO v5	<u>1.52</u>	<u>10.25</u>	84.45	82.71
Raw Qwen3 baselines (frozen project protocol)				
Qwen3-4B-Base (raw)	33.03	61.59	92.01	120.33
Qwen3-4B-Instruct (raw)	4.01	23.42	<u>59.80</u>	70.17
Historical external baselines				
VietAI	–	–	–	–
GPT-4o	8	11	187	58
GPT-3.5-Turbo	18	17	**47**	**30**
Qwen3-14B	50	45	131	<u>34</u>
Llama3.3-70B-Instruct	14	13	186	73
Phi4-14B	205	214	166	134
Sailor-20B-chat	38	43	268	41
VinBigdata (7B)	115	89	98	82

Table 4: Absolute summary length distance across the four report datasets. Lower is better. Bold marks the best reported value in each column and underlining marks the second-best reported value across both project and baseline rows. The two final systems and the raw Qwen3 baseline rows use the frozen project protocol; the remaining external rows are contextual historical comparisons only. ViMs and VLSP have split overlap.

nal rows in the main benchmark rather than collapsing the method to one backbone only.

5.3 Within-project Stage-wise Ablation

Table 5 compresses the internal project comparison into one stage-wise ablation over VietNews and WikiLingua. Its purpose is to analyze the observed effect of the two-stage post-training pipeline (SFT + GRPO v5) relative to SFT-only, pretrained-only, and fresh-GRPO variants. Appendix A.1 provides the full four-dataset within-project tables, while Appendix A.2 records the historical v4-v5 bundle comparison that motivated the final GRPO choice. To better visualize this multi-stage interaction on the primary dual objectivesretaining content quality while enforcing a strict length budgetwe plot the joint ROUGE-2 and absolute summary length distance trajectory in Figure 3.

The stage-wise comparison yields three observations about the evaluated training recipe:

SFT provides the largest stage-wise gain. Both pretrained backbones have substantially lower ROUGE-2 than their SFT variants on both single-document datasets. The largest observed gains occur after SFT, making it the dominant improvement stage in the present comparison. The subsequent GRPO results are therefore interpreted relative to this strong SFT baseline. GRPO v5 then acts as an *alignment refiner* in the evaluated systems. It improves ROUGE-2 for both SFT-initialized backbones. For the Base backbone it also reduces length distance on both single-document datasets; for the Instruct backbone, length distance remains close to SFT-only and is slightly worse. For instance, on the Base backbone, GRPO refinement reduces length deviation on VietNews by 80% (from 5.73 to 1.16 words), while ROUGE-2 rises from 26.61 to 28.97.

Fresh GRPO underperforms SFT initialization. Applying GRPO directly to the pretrained checkpoint (Fresh GRPO v5, without the intermediate SFT step) underperforms SFT-initialized GRPO on both single-document datasets. As shown by the isolated markers in Figure 3, the Fresh GRPO models reach only 20.73 and 18.67 ROUGE-2 on VietNews, compared with 28.97 and 29.62 after SFT initialization. This result establishes the empirical importance of SFT warm-starting in the present setup. The multiplicative quality gate is a plausible contributing mechanism, but the current experiments do not isolate it causally.

Observed trade-offs and backbone choice. The trajectory curves in Figure 3 visualize the

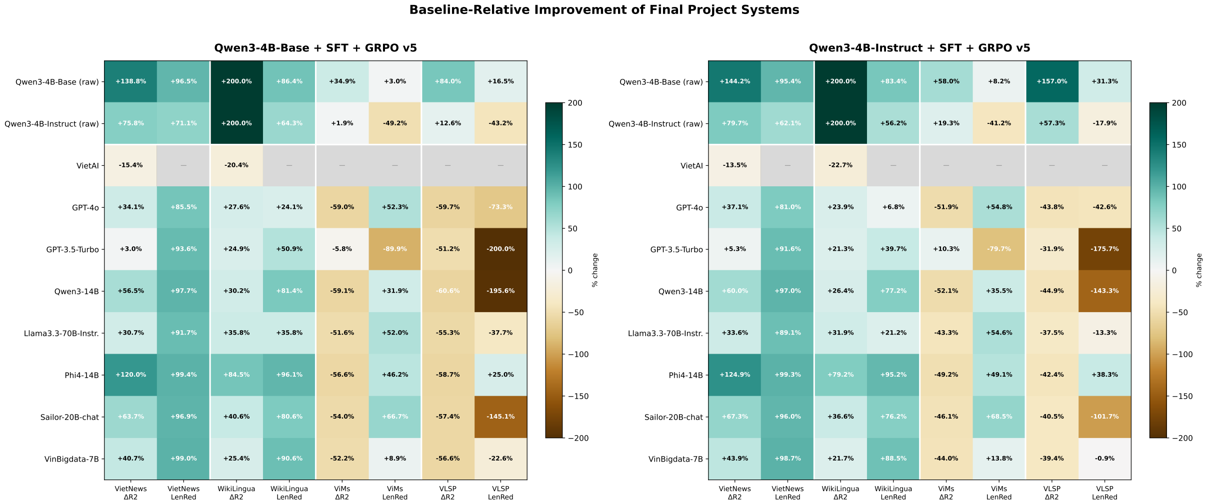

Figure 2: Baseline-relative improvement summary for the two final project systems. Each dataset is shown explicitly as a two-column block: positive ΔR2 means higher ROUGE-2, positive LenRed means smaller absolute summary length distance, and gray cells mark unavailable values. The first two rows are raw Qwen3-4B baselines, followed by external baselines. Cell text shows the exact percentage change, while color intensity is clipped for readability.

Model	VietNews ROUGE-2	WikiLingua ROUGE-2	VietNews LenDist	WikiLingua LenDist
Base pretrained	12.13	3.18	33.03	61.59
Base + SFT	26.61	24.43	5.73	14.18
Base + fresh GRPO v5	20.73	9.71	3.80	22.03
Base + SFT + GRPO v5	<u>28.97</u>	**26.35**	<u>1.16</u>	**8.35**
Instruct pretrained	16.48	7.44	4.01	23.42
Instruct + SFT	26.70	23.93	**1.09**	<u>9.87</u>
Instruct + fresh GRPO v5	18.67	8.09	2.95	16.12
Instruct + SFT + GRPO v5	**29.62**	<u>25.59</u>	1.52	10.25

Table 5: Compact within-project ablation on VietNews and WikiLingua. This table is included to explain the chosen training recipe, not as the paper's headline leaderboard.

two-stage warm-start as a clear shift towards the top-right corner. For the Base branch, the absolute length distance decreases by 89.9% on the macro average (47.31 words to 4.76 words) coupled with a +261.3% relative gain in macro ROUGE-2. For the Instruct branch, macro length distance is slashed by 57.1% (13.72 to 5.89 words) alongside a +130.8% relative macro ROUGE-2 gain. Among the evaluated systems, the Instruct-initialized pipeline has the highest VietNews ROUGE-2 (29.62), whereas the Base-initialized pipeline has the lowest single-dataset length distance (1.16 words). Together, these results show consistent gains from the two-stage recipe across the two backbone families evaluated here.

5.4 Reward-hacking Mitigation Analysis

Reward hacking occurs when a model optimizes the reward signal without improving output quality. In v3 (the pre-safeguard bundle), three interacting design flaws created conditions under which this behavior emerged. First, v3 used a **linear reward sum**

$$R_{\text{total}} = w_{\text{acc}} R_{\text{acc}} + w_{\text{len}} R_{\text{len}} + w_{\text{sent}} R_{\text{sent}},$$

which allowed the constraint terms to contribute independently of content: a model could score ≈ 0.50 by satisfying length and sentence budgets with garbage while ignoring R_{acc}. Second, a repetition penalty of 1.3 was applied by the HuggingFace decoder over the full $\approx$ 2,000-token input sequence, penalizing the model for reusing any vocabulary from the source article and pushing generation toward out-of-distribution symbols. Third, n-gram blocking with size 3 blocked source trigrams, forbidding named entities and key Vietnamese content words. The resulting failure modes, concentrated in the Base branches where the pretrained backbone lacks the instruction-following prior of the Instruct model, included repetition loops (e.g., `assistant` repeated 128 times, ROUGE-2 = 0.0), punctuation

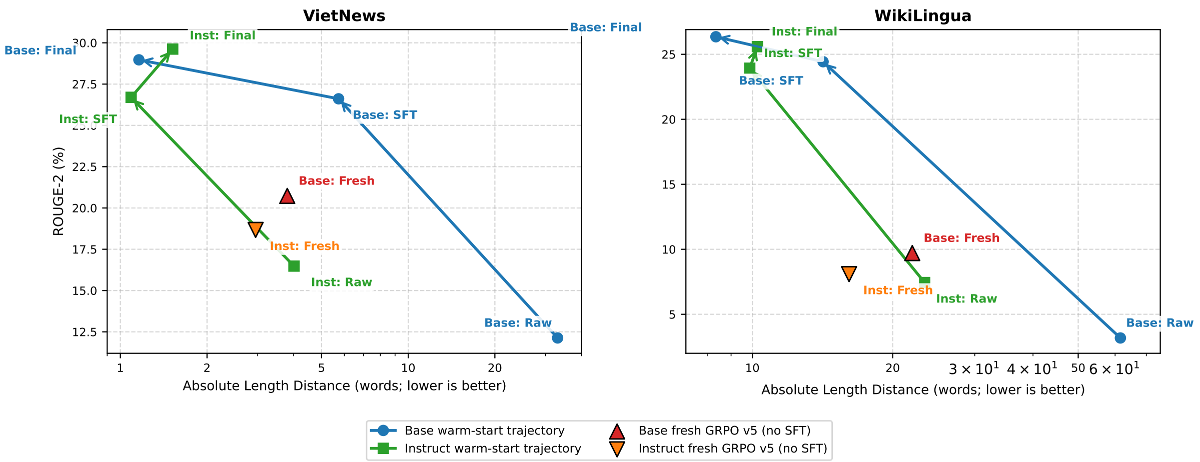

Figure 3: Two-panel trade-off trajectory showing ROUGE-2 versus absolute summary length distance for the Base and Instruct backbones on VietNews (left) and WikiLingua (right). Arrowed lines display the observed stage sequence (Raw $\rightarrow$ SFT $\rightarrow$ SFT+GRPO v5), while standalone markers show the lower content scores of Fresh GRPO v5 without SFT initialization.

padding (245 words of periods, LenErr $= 1189\%$), digit blobs that satisfied "khong qua X tu" with $R_{\text{len}} = 1.0$ and $R_{\text{acc}} = 0$, and comma-separated token loops carrying no semantic content. Figure 4 illustrates the two most severe modes qualitatively. Appendix A.2 records the historical bundle-level v4-v5 comparison, and Appendix A.7 keeps the sentence-reward ablation that remains inconclusive under the frozen sampling protocol.

The canonical v5 bundle addresses each root cause through the defense-in-depth safeguards described in Section 4.4: the generation-layer fixes restore natural decoding, the detection-layer filter catches degenerate outputs, and the multiplicative gate structurally prevents constraint-only optimization.

Table 6 reports the quantitative evidence. Base fresh shows the most dramatic recovery: degenerate rate collapses from 23.81% to 0.32% ($\approx 74\times$ reduction), $R_{\text{acc}} = 0$ drops from 27.16% to 1.94%, and ROUGE-2 rises from 10.58 to 18.66. Base SFT-init suffered severe length explosion under v3 (LenErr $= 425\%$), which v5 cuts to 14.94% while raising ROUGE-2 from 18.03 to 26.38. The Instruct branches were substantially more robust under v3—degenerate rates remained below 1%—owing to their chat-tuning prior; their v3-to-v5 gains are correspondingly incremental, with Instruct SFT-init v5 reaching the highest mean R_{acc} (0.4213) and ROUGE-2 (27.65). Across all eight configurations, the v5 safeguards eliminate catastrophic failure modes without harming any branch, and the recovery magnitude scales inversely with v3 baseline quality.

These results establish v5 as the stable canonical GRPO configuration. The Instruct branches' near-immunity to degeneration reinforces the finding from Section 5.3 that an SFT warm-start or equivalent generative prior is a prerequisite for effective GRPO in this setting. The v3-to-v5 improvement is properly understood as the effect of a **mitigation package**—generation, detection, and formula-level safeguards acting jointly—rather than an ablation of any single component. Appendix A.4 retains the sentence-control results that justify keeping sentence reward as an explicit metric, and Appendix A.7 shows that removing that term does not support a clean causal claim under the current evaluation protocol.

6 Conclusion

Controllable Vietnamese summarization is not only a content-generation problem, but also an alignment problem: the model must preserve salient information while obeying explicit budgets on summary length and sentence count. This paper studies that problem in a lightweight post-training setting for Qwen3-4B models, with the goal of identifying a practical recipe that improves controllability without full-model retraining.

The resulting method is a two-stage pipeline. Stage 1 uses instruction-conditioned LoRA-SFT to teach summarization behavior together with control-aware instruction following, and Stage 2 applies GRPO alignment to refine policy behavior under those constraints. The main contribution of

Run	ROUGE-2	LenErr%	Deg.%	$R_{\text{acc}} = 0\%$	Mean R_{acc}
Base fresh v3	10.58	163.16	23.81	27.16	0.1854
Base fresh v5	18.66	29.05	0.32	1.94	0.3160
Base SFT-init v3	18.03	425.00	14.77	1.16	0.2508
Base SFT-init v5	<u>26.38</u>	**14.94**	2.71	**0.45**	<u>0.4009</u>
Instruct fresh v3	15.00	30.08	**0.03**	1.48	0.2768
Instruct fresh v5	16.69	22.78	<u>0.03</u>	1.58	0.2979
Instruct SFT-init v3	25.06	19.94	0.94	1.19	0.3837
Instruct SFT-init v5	**27.65**	<u>15.85</u>	1.42	<u>0.45</u>	**0.4213**

Table 6: Reward-hacking mitigation from v3 to v5, computed from stored generations with the current reward implementation. Deg.% is the fraction flagged as degenerate by the detection filter; $R_{\text{acc}} = 0\%$ is the fraction receiving zero content reward; Mean R_{acc} is the average content reward. The comparison is a package-level ablation and does not isolate any single change.

the second stage is not only the PPO-style policy update, but the full reward-and-safeguard package around it: a gated reward that makes content quality a prerequisite for constraint reward, prompt-aware length and sentence rewards, and defense-in-depth anti-hacking measures spanning generation settings, degeneracy detection, and formula-level protection.

Across VietNews and WikiLingua, this recipe improves substantially over the raw Qwen3 backbones and clarifies the role of each stage. SFT provides the largest quality gain, while SFT-initialized GRPO further improves the ROUGE-length trade-off; by contrast, fresh GRPO remains clearly weaker than warm-started GRPO. In the final benchmark, the Instruct pipeline achieves the highest VietNews ROUGE-2 (29.62), while the Base pipeline yields the strongest length control on the primary single-document datasets, reaching absolute length distances of 1.16 on VietNews and 8.35 on WikiLingua. The reward-hacking analysis adds a second practical insight: GRPO is effective in this setting only when coupled with an appropriate reward design, because the v5 safeguard bundle sharply reduces degenerate outputs and stabilizes the previously fragile branches.

The paper's conclusions are intentionally bounded. Strong single-document claims are supported only on VietNews and WikiLingua; ViMs and VLSP remain diagnostic only because of split overlap. Their weaker results are also plausible under the present training recipe itself: multi-document supervision is severely underrepresented, contributes little to the dominant SFT stage, and is optimized with flat cluster serialization plus a single-reference ROUGE-oriented reward that is better matched to single-document control than to cross-document fusion. The external baseline rows are useful context rather than protocol-matched reruns, and the frozen evaluation protocol still relies on sampling without a saved seed. Moreover, the present experiments do not causally isolate the contribution of each safeguard component or each hyperparameter inside the v3, v4, and v5 GRPO bundles.

Future work should therefore strengthen multi-document coverage, evaluation, and causal analysis as much as model capability. The most important next steps are to build leakage-free multi-document testbeds, expand and rebalance multi-document supervision during both SFT and GRPO, compare flat cluster serialization with more structure-aware encodings or hierarchical fusion mechanisms, rerun competitive baselines under a shared protocol, save deterministic evaluation seeds, and conduct targeted ablations that separate initialization choice, reward terms, safeguard layers, and GRPO configuration effects. With those extensions, the pipeline developed here can serve as a stronger foundation for controllable Vietnamese summarization beyond the present single-document setting.

Case	Reference	v3 output (failure)	v5 output (recovered)
Base fresh idx 14	Điều tra vụ anh rể dùng dao đâm chết em vợ vào rạng sáng	**Output** *assistant assistant assistant assistant assistant assistant … (×128 tokens)* **Failure signature** • Pure repetition loop — no content • ROUGE-2 = 0.00 • LenErr = 814.29% • R_acc = 0.00	**Output** *Anh C. bị anh rể đâm chết tại nhà.* **Recovered properties** • Short, content-bearing headline • Named entity retained • Budget restored • ROUGE-2 = 0.19
Base SFT-init idx 7	5 năm chưa xử xong vụ án "lạ" có hơn 30 luật sư bào chữa miễn phí	**Output** *Bị cáo Tuyết bị cấp sơ thẩm 2 lần tuyên phạt 12 năm tù . Bị cáo Tuyết kháng cáo , nên 2 lần vụ án được đưa ra xét xử phúc thẩm và đều có chung 1 kết quả , huỷ án . . . .* **Failure signature** • 245 words — punctuation-heavy blow-up • Meaningful start → punctuation flood • LenErr = 1189.47% • ROUGE-2 = 0.034	**Output** *Vụ án "lạ" : Huỷ án 2 lần , bị cáo vẫn bị tạm giam 5 năm !* **Recovered properties** • Compact, accurate rewrite • Budget restored • Content preserved • ROUGE-2 = 0.22

Figure 4: Paired v3–v5 examples of the two most severe reward-hacking modes from VietNews. Each card shows a reference summary (gray), the failed v3 output (red), and the recovered v5 output (green), together with failure or recovery indicators. Left: repetition loop (Base fresh; `assistant` repeated 128 times). Right: punctuation-driven length explosion (Base SFT-init; 245 words of periods).

A Appendix

A.1 Full Within-project Variant Tables

Tables 7–9 provide the full four-dataset view for all canonical Qwen3 project variants. These tables are retained as internal context, but they are not the headline benchmark of the paper.

Model	VietNews	WikiLingua	ViMs	VLSP
Base pretrained	12.13	3.18	13.46	9.49
Base + SFT	26.61	24.43	**25.65**	**28.14**
Base + fresh GRPO v5	20.73	9.71	20.28	18.14
Base + SFT + GRPO v5	<u>28.97</u>	**26.35**	18.16	17.46
Instruct pretrained	16.48	7.44	17.83	15.51
Instruct + SFT	26.70	23.93	<u>24.71</u>	<u>27.03</u>
Instruct + fresh GRPO v5	18.67	8.09	18.43	16.05
Instruct + SFT + GRPO v5	**29.62**	<u>25.59</u>	21.27	24.39

Table 7: ROUGE-2 (%) across all four datasets for the canonical Qwen3 project variants.

Model	VietNews	WikiLingua	ViMs	VLSP
Base pretrained	33.03	61.59	92.01	120.33
Base + SFT	5.73	14.18	66.45	<u>61.64</u>
Base + fresh GRPO v5	3.80	22.03	71.77	88.02
Base + SFT + GRPO v5	<u>1.16</u>	**8.35**	89.23	100.50
Instruct pretrained	4.01	23.42	**59.80**	70.17
Instruct + SFT	**1.09**	<u>9.87</u>	64.55	**62.22**
Instruct + fresh GRPO v5	2.95	16.12	<u>63.05</u>	72.91
Instruct + SFT + GRPO v5	1.52	10.25	84.45	82.71

Table 8: Absolute summary length distance across all four datasets for the canonical Qwen3 project variants.

Model	VietNews	WikiLingua	ViMs	VLSP
Base pretrained	205.89	174.67	42.61	47.93
Base + SFT	31.19	24.72	38.69	27.53
Base + fresh GRPO v5	22.36	50.94	31.89	34.33
Base + SFT + GRPO v5	<u>6.43</u>	**14.69**	44.42	42.55
Instruct pretrained	24.50	57.47	**26.07**	<u>27.09</u>
Instruct + SFT	**6.26**	18.34	31.55	**24.51**
Instruct + fresh GRPO v5	17.08	38.67	<u>28.37</u>	28.76
Instruct + SFT + GRPO v5	9.24	<u>18.11</u>	39.14	32.85

Table 9: Relative length error (%) across all four datasets for the canonical Qwen3 project variants.

A.2 Historical Comparison of Two GRPO Configuration Bundles

This appendix records the two GRPO configuration bundles available in the stored artefacts: v4 ($K = 4$, learning rate 5×10^{-7}, $\beta = 0.15$) and v5 ($K = 8$, learning rate 2×10^{-6}, $\beta = 0.04$). Because only two operating points were evaluated, three hyperparameters change jointly, and the sampling-based evaluation has no saved seed, this comparison is not a controlled hyperparameter ablation or a systematic search.

Branch	v4 ROUGE-2	v5 ROUGE-2	v4 LenErr%	v5 LenErr%	v4 LenDist	v5 LenDist
Base fresh	10.93	15.22	91.67	36.65	26.70	12.90
Base SFT-init	26.25	27.66	17.88	10.56	7.05	4.80
Instruct fresh	12.39	13.38	33.98	27.88	11.65	9.50
Instruct SFT-init	26.23	27.60	13.09	13.68	6.00	5.85

Table 10: Historical comparison of two explored GRPO configuration bundles on VietNews and WikiLingua macro averages. These observations do not identify the effect of any individual hyperparameter or establish a globally optimal configuration.

Across the stored runs, v5 has higher ROUGE-2 in all four branches and lower length distance in all four branches. Relative length error is lower under v5 in three branches and slightly higher for Instruct SFT-init. These values are retained for transparency and hypothesis generation only, not as evidence for a causal configuration claim.

A.3 Multi-document Baseline Comparison

Table 11 keeps the full baseline-relative comparison on ViMs and VLSP. It is retained in the appendix because it is useful diagnostic context, but it should not dominate the main method-first narrative of the paper.

Baseline	Qwen3-4B-Base + SFT + GRPO v5		Qwen3-4B-Instruct + SFT + GRPO v5	
	Δ ROUGE-2$\uparrow$	LenDist reduction$\uparrow$	Δ ROUGE-2$\uparrow$	LenDist reduction$\uparrow$
VietAI	–	–	–	–
GPT-4o	-59.4%	+22.6%	-47.9%	+31.8%
GPT-3.5-Turbo	**-35.3%**	-146.4%	**-17.1%**	-117.1%
Qwen3-14B	-59.8%	-15.0%	-48.5%	-1.3%
Llama3.3-70B-Instruct	<u>-53.5%</u>	+26.7%	<u>-40.4%</u>	+35.5%
Phi4-14B	-57.7%	<u>+36.8%</u>	-45.7%	<u>+44.3%</u>
Sailor-20B-chat	-55.7%	**+38.6%**	-43.2%	**+45.9%**
VinBigdata (7B)	-54.5%	-5.4%	-41.6%	+7.1%

Table 11: Relative change against each contextual baseline on the ViMs and VLSP macro average. Positive values are better for both columns, but the table is diagnostic only because both datasets overlap with project training or validation pools.

A.4 Sentence-control Results

Sentence-control results are retained here because the main paper focuses on ROUGE-2 and absolute summary length distance.

Model	VN Exact%	VN Tol.%	VN MAE	WL Exact%	WL Tol.%	WL MAE
Base pretrained	73.95	89.55	0.550	15.4	32.4	2.910
Base + SFT	91.40	96.55	4.299	58.6	83.8	1.674
Base + fresh GRPO v5	91.80	99.30	0.089	34.2	72.8	1.226
Base + SFT + GRPO v5	93.85	98.75	0.316	65.6	<u>95.6</u>	0.470
Instruct pretrained	93.95	<u>99.40</u>	0.067	34.2	64.6	1.582
Instruct + SFT	<u>95.45</u>	98.90	0.058	**88.0**	97.0	0.204
Instruct + fresh GRPO v5	94.85	**99.60**	<u>0.056</u>	46.8	72.4	1.178
Instruct + SFT + GRPO v5	**96.95**	99.05	**0.041**	<u>78.8</u>	95.6	<u>0.340</u>

Table 12: Sentence-control metrics recomputed from stored generations. "Tol." counts a prediction as correct when the sentence count is within ±1 of the requested budget.

A.5 Dataset Processing Details

The final JSONL artefacts are produced by the project's data processing and augmentation modules. The most important preprocessing rules are listed below.

- Shared preprocessing truncates sources to 8,000 characters and summaries to 1,500 characters, then replaces underscore-based segmentation markers with spaces.
- VietNews uses the first line of each file as the target headline, the sapo plus body as source text, and drops the last line as caption-like trailing content.
- WikiLingua flattens the source and target sentence lists into single strings.
- ViMs reads each document in a cluster and concatenates the documents into a single source with explicit boundary markers.
- VLSP flattens each cluster in the same style, prefers the abstractive summary field, and falls back to label-based extraction only when a summary is unavailable.
- SFT and GRPO intentionally use different training pools: SFT filters short VietNews titles and excludes VLSP, while GRPO keeps all VietNews and includes VLSP.

A.6 Pipeline Configurations and Prompt Templates

This subsection summarizes the end-to-end pipeline configuration used by the paper. The source of truth is the project's canonical configuration files, training scripts, evaluation pipeline, and audit documentation. Prompt examples are shown in an ASCII-friendly transliteration for toolchain compatibility, but they follow the same instruction format used throughout the pipeline.

Process	Prompt shape	Template policy	Stored outputs
SFT	system + user + assistant	Always uses the approximate control template family	messages plus metadata
GRPO train/val	system + user	Rotates three length templates and three sentence templates across prompts	prompt, reference, and metadata
Evaluation/test	system + user	Deterministic test prompt; template index fixed to 0	prompt, reference, and metadata

Table 13: Prompt-construction rules used by the pipeline.

Signal	Template families	Notes
Length	*khoang X tu*; *trong khoang lo-hi tu*; *khong qua X tu*	SFT uses only the first form; GRPO rotates all three; eval fixes the first form
Sentence	*khoang X cau*; *trong khoang lo-hi cau*; *khong qua X cau*	Same rotation policy as the length templates

Table 14: Control-template families used in the pipeline.

Data augmentation and prompt construction. All prompt-bearing samples preserve metadata for the reward functions and analysis logs. The core metadata fields are target length, target sentence count, length requirement text, and sentence requirement text; the pipeline also tracks statistics such as source length.

Setting	Canonical paper configuration
Backbone family	Qwen3-4B-Base or Qwen3-4B
LoRA adapter	$r = 32$, $\alpha = 32$, dropout 0.05; attention projections (query, key, value, output) and MLP projections (gate, up, down)
Sequence length	3072 on H200/A100; 2048 on V100
Training	$lr = 5 \times 10^{-5}$, cosine scheduler, $warmup\_ratio = 0.1$, $epochs = 2.0$
Thinking suppression	Enabled
Batch calibration	Enabled before training
Checkpoint behavior	Model-family-specific output directory; resume from latest checkpoint when present

Table 15: Canonical SFT configuration used by the paper.

GPU tier	Start batch	Grad. accum.	Effective batch
H200	10	2	20
A100	6	3	18
V100	2	4	8

Table 16: SFT per-GPU starting presets used in the canonical training runs. Auto-calibration can adjust these values at runtime.

Canonical SFT configuration. The appendix tables intentionally follow the launch-script presets rather than the raw library defaults, because the launch scripts are the source of truth for the canonical paper runs.

Setting	Canonical paper configuration
Initialization	SFT-initialized warm start
Core v5 hyperparameters	$K = 8$, $lr = 2 \times 10^{-6}$, $\beta = 0.04$, $total\_steps = 800$
Rollout decoding	Temperature 0.7, top-p 0.9, random sampling, max 80 new tokens, max sequence length 3072
Reward weights	Accuracy 0.5, length 0.3, sentence 0.2
Stability settings	Gradient checkpointing enabled, thinking mode suppressed, auto-calibration enabled
Repetition controls	Repetition penalty 1.0, n-gram blocking disabled

Table 17: Canonical GRPO v5 configuration used in the main paper results.

GPU tier	Start batch	Grad. accum.	Effective prompts/update
H200	4	4	16
A100	2	8	16
V100	1	16	16

Table 18: GRPO per-GPU starting presets used in the canonical training runs. Auto-calibration can refine these values at runtime.

Canonical GRPO v5 configuration. For reference only, the main ablation bundle changes are *v4*: $K = 4$, $lr = 5 \times 10^{-7}$, $\beta = 0.15$; *v3* and *v4* remain ablation settings and should not be read as the canonical appendix configuration.

Setting	Frozen evaluation protocol
Decoding	Temperature 0.3, top-p 0.9, random sampling, max 256 new tokens
Batching	Batch size 8
Prompt formatting	Chat template applied with thinking mode disabled when supported
Prompt truncation	Tokenizer truncates prompts to 3072 minus the maximum allowed new tokens
Evaluation inputs	Per-dataset test artefacts plus the combined test split
Metrics in main paper	ROUGE-2 and Length Distance; sentence-control kept in appendix
Randomness	One-shot low-temperature sampling; no saved seed

Table 19: Frozen evaluation configuration used to generate the paper tables.

Evaluation configuration. **SFT prompt example.**

System: Ban la mot tro ly AI chuyen tom tat van ban tieng Viet. Hay tao ra ban tom tat ngan gon, chinh xac, tuan thu dung yeu cau ve do dai va so cau.
User:

- Do dai: khoang 22 tu
- So cau: khoang 1 cau

Van ban: <source article>
Assistant: <gold summary>

GRPO rollout prompt example.

System: Ban la mot tro ly AI chuyen tom tat van ban tieng Viet. Hay tao ra ban tom tat ngan gon, chinh xac, tuan thu dung yeu cau ve do dai va so cau.
User:

- Do dai: trong khoang 18–26 tu
- So cau: trong khoang 1–2 cau

Van ban: <source article>
reference = <gold summary>
meta.target_length = 22
meta.target_sentences = 1
meta.length_requirement = "trong khoang 18–26 tu"
meta.sentence_requirement = "trong khoang 1–2 cau"

Evaluation/test prompt example.

System: Ban la mot tro ly AI chuyen tom tat van ban tieng Viet. Hay tao ra ban tom tat ngan gon, chinh xac, tuan thu dung yeu cau ve do dai va so cau.
User:

- Do dai: khoang 22 tu
- So cau: khoang 1 cau

Van ban: <source article>
reference = <gold summary>
meta.dataset = "vietnews"

GRPO training and evaluation share the same prompt shape: both keep only the system and user turns, store the gold summary separately as *reference*, and preserve the control metadata in *meta*. The differences are that GRPO rotates all template families during rollout training, whereas evaluation fixes the deterministic approximate template, and SFT is the only stage that includes an assistant turn inside the serialized chat sample.

A.7 No-sentence-reward Ablation

The no-sentence ablation is informative but not decisive. Both sides of the comparison still use sampling without a saved evaluation seed, so the correct conclusion is *inconclusive* rather than a clean causal claim about sentence reward.

Branch	w_{sent}	ROUGE-2	LenErr%	Sent. Exact%	Sent. MAE
Base fresh v4	0.2	13.88	76.44	59.42	**1.136**
Base fresh ablation	0.0	13.99	76.50	60.32	<u>1.163</u>
Base SFT-init v4	0.2	<u>26.71</u>	<u>20.82</u>	<u>73.39</u>	2.009
Base SFT-init ablation	0.0	**26.85**	**17.21**	**73.65**	1.452

Table 20: No-sentence-reward ablation computed from a frozen evaluation snapshot. The differences are too small under the current sampling-based protocol to support a clean causal statement.